

Erdős Problem 915 — Cheat Sheet

the sixteen variants, one line per bound

PROVED

exact value established

PARTIAL

partial or conjectural

OPEN

general value open

Simple graphs

$\ell_m(n)$ edge

PROVED

UP $\lfloor \frac{m(n-1)}{2} \rfloor$, $n \geq m \geq 2$ — Gomory–Hu tree, double-counted against Mader’s degree bound

LO same formula — hub-and-spokes graph, spoke degree $\leq m - 2$

$k_m(n)$ vertex

PARTIAL

UP $= \ell_m(n)$ $m \leq 4$ — edge bound does not transfer ($\kappa \leq \lambda$ is the wrong direction), equal anyway up to $m = 4$

LO $\lfloor \frac{m(n-1)}{2} \rfloor$ — same hub-and-spokes; Sørensen–Thomassen overtakes it from $m = 5$, $n \geq 14$

$\ell_m^{\text{dir}}(n)$ arc

OPEN

UP $M(n) = \max(2(n-1), \lfloor n^2/4 \rfloor)$ $m = 2$ — reachability skeleton (at $m = 2$ a feasible digraph is its own skeleton)

LO $M(n)$ — bidirected hub / one-way bipartite digraph

$k_m^{\text{dir}}(n)$ vertex

OPEN

UP $M(n)$ $m = 2$ — skeleton routes are already vertex-disjoint, so vertex bound comes first, arc bound follows

LO $\lfloor n^2/4 \rfloor$ — one-way bipartite digraph

Multigraphs

$L_m(n)$ edge

PROVED

UP $(m-1)(n-1)$, every n, m — Gomory–Hu, $n-1$ cuts each of capacity $\leq m-1$

LO same formula — any spanning tree at multiplicity $m-1$

$K_m(n)$ vertex

PARTIAL

UP $2(n-1)$ $m = 3$ — $\kappa(u, v)$ = copies of the edge + disjoint detours, capped together

LO $(m-1)(n-1)$ — doubled tree: no detours, so the whole cap goes to multiplicity

$L_m^{\text{dir}}(n)$ arc

PROVED

UP $(m-1)M(n)$, every n, m — peel a reachability skeleton off $m-1$ times

LO same formula — thickened bidirected tree / one-way bipartite digraph

$K_m^{\text{dir}}(n)$ vertex

OPEN

UP $(m-1) \lfloor n^2/4 \rfloor + 4(m-1)^2(n-1)$ not tight

LO $n(n-1)(m+1-n)$ proposed, $2 \leq n \leq m$ — thickened complete digraph, unique isomorphism class at $n = 3, m = 6$

Hypergraphs ($r = 3$)

$\ell_m^{(r)}(n)$ edge

PARTIAL

UP $\lfloor \frac{(m-1)(n-1)}{r-1} \rfloor$ — cut induction on trace hypergraphs (Berge forbids the graph reduction)

LO same formula $m-1 \leq \binom{n-2}{r-2}$ — star hypertree / degree-capped hub

$k_m^{(r)}(n)$ vertex

PARTIAL

UP $\lfloor \frac{n-1}{r-1} \rfloor$ $m = 2$ — incidence graph is a forest ($r|E| \leq n + |E| - 1$)

LO $\lfloor \frac{(m-1)(n-1)}{r-1} \rfloor$ $r \geq 3, m-1 \leq \binom{n-2}{r-2}$ — star hypertree

$\ell_m^{\text{dir}(r)}(n)$ arc

OPEN

UP $\frac{(m-1)n(n-1)}{r-1}$ not tight — tail-head shadow digraph

LO $n \cdot \min(m-1, \binom{n-1}{r-1})$ $n \geq r$ — bounded outgoing hyperedge count

$k_m^{\text{dir}(r)}(n)$ vertex

OPEN

UP same as arc

LO same as arc — outgoing cut never looks at interior vertices, one argument serves both

Multihypergraphs ($r = 3$)

$L_m^{(r)}(n)$ edge

PARTIAL

UP $\lfloor \frac{(m-1)(n-1)}{r-1} \rfloor$, unchanged by repetition — same cut induction, repetition-blind

LO $\frac{(m-1)(n-1)}{r-1}$ $(r-1) \mid (n-1)$ — star hypertree, every hyperedge repeated $m-1$ times

$K_m^{(r)}(n)$ vertex

PARTIAL

UP $\lfloor \frac{2(n-1)}{r-1} \rfloor$ $m = 3$ — incidence-graph cycle rank via cut vertices / separating pairs

LO $\frac{(m-1)(n-1)}{r-1}$ $(r-1) \mid (n-1)$ — repeated star hypertree

$L_m^{\text{dir}(r)}(n)$ arc

OPEN

UP $\frac{(m-1)n(n-1)}{r-1}$ — tail-head pair counting

LO $n(m-1)$ $n \geq r$ — every tail gets $m-1$ copies; meets the bound exactly at $n = r$

$K_m^{\text{dir}(r)}(n)$ vertex

OPEN

UP same as arc

LO same as arc — general (any tail/head split) orientation shares the same constant